ASILAH MARYAM

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I am a Mechanical Engineer with a specialization in robotics who creates things that move, help, and inspire. I am driven by the intersection of innovation, sustainability, and impact, evident through my experience building rovers for Mars-like environments and developing systems that save people's lives,

SKILLS

CAD/Analysis: SolidWorks, Engineering Drawings, Finite Element Analysis (FEA), GD&T (Geometric Dimensioning & Tolerancing
Programming: MATLAB, Arduino, Java, Python (data analysis), HTML, CSS
Manufacturing/Prototyping: Machining (lathe, mill), Automation, Rapid Prototyping (3D Printing)
Leadership/Other: Team Leadership, Project Management, Web Design, Graphic Design, Tableau
Languages: English (Native), Urdu/Hindi (Conversational), Spanish (Conversational)

PROFESSIONAL EXPERIENCE

Intuitive Surgical, Sunnyvale, CA — June 2025 - September 2025
R&D Mechanical Engineering Intern (Multiport Endoscope Team)

- Evaluated pressure equalization methods by designing and manufacturing a Device Under Test (DUT) utilizing GD&T principles and running autoclave condition tests.
- Completed a worst-case and RSS tolerance stackup to determine and mitigate critical sources of variation in button actuation.
- Developed a comprehensive test plan and performed data analysis, concluding with actionable design suggestions for the team.

Amazon Lab 126, Sunnyvale, CA — June 2024 - September 2024
Hardware Dev Engr Intern (Prototyping, Automation, and Calibration Team)

- Engineered a suspended automation system using a 4 Cable Driven Parallel Robot (CDPR) configuration to provide high flexibility in critical product testing processes.
- Performed mechanical system analysis using MATLAB to ensure structural integrity and optimize performance for a 15lb payload.
- Wrote algorithms and programmed components via Arduino IDE , incorporating advanced mathematical and geometric functions to improve payload speed and path efficiency.
- Engaged with 3 external suppliers to source servos, pulleys, and belts based on motor and mechanical advantage specifications.

Amazon Lab 126, Sunnyvale, CA — June 2023 - September 2023
AFE Hardware Intern (Prototyping, Automation, and Calibration Team)

- Developed a cost-effective automated button-pushing system within $250 for testing Amazon Fire TV and competitor remotes, providing an alternative to reworking remotes.
- The system researched, devised, and built reduced the required time for rework from 2 weeks to a couple of hours.
- Researched and evaluated 4 actuators to identify the optimal solution, focusing on cost, durability, and compatibility.
- Utilized rapid prototyping through SolidWorks to design and iterate the button pusher mechanism, ensuring compatibility with existing testing infrastructure.

LEADERSHIP

urc@ucla, Los Angeles, CA — February 2024 - Present
Co-founder and Education Lead

- Create workshops for members to learn how to design, build, wire, and program robotics systems to prepare for the 2025 and 2026 University Rover Challenges

- Co-lead an all women's robotics team of over 40 members in design, manufacturing, programming, electrical, science, and marketing sub-teams

PROJECTS

University Rover Challenge, Los Angeles, CA — February 2024 - Present
Team
- Manufactured arm should and elbow, suspension, and chassis
- Wired and programmed motors and linear actuators and set up communication system

Suspended Automation Mechanism, Sunnyvale, CA — June 2024 - September 2024
Individual Intern Project
- Created a 4 end cable driven parallel robot carrying a payload of 15lbs in the X, Y, and Z planes

3lb Combat Robot, Los Angeles, CA — October 2023 - May 2024
Team
- Manufactured a 3lb ring spinner combat robot winning "King of the Ring"

Boximizer, Los Angeles, CA — January 2024
Individual Intern Project
- Prototyped a device in 1.5 days for IDEA Hacks focused on cutting down excess waste generated from oversized packages by custom sizing and cutting a box based on an object's dimension

Low Cost Button Pusher, Sunnyvale, CA — June 2023 - September 2023
Individual Intern Project
- Researched, devised, and built a cost-effective automated button pusher for testing Fire TV remotes to reduce 2 week time required for rework to a couple of hour

Bluetooth Speaker, Sunnyvale, CA — June 2023
Individual Project
- Designed and assembled a bluetooth speaker with 3 modular units capable of stacking atop each other and spinning in different directions to enhance audio coverage over a wider area than traditional front facing speaker

Underwater Robot, Los Angeles, CA — September 2022 - December 2022
Team Project
- Designed, 3D modeled, and wired a rover to “swim” underwater with joystick control

Cake Wagon, Fremont, CA — November 2021
Individual Project
- Designed, built, baked, decorated, and programmed a remote controlled cake